MODULE 05

Time Series and Predictive Modeling

DATA SCIENCE · LEARNING NOTES

Mohan Babu University · Course-aligned study guide

Concepts • examples • practice • revision

01. Importing and cleaning

Learning focus. This section introduces importing and cleaning as part of time series and predictive modeling. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to importing and cleaning. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

02. Plotting time series

Learning focus. This section introduces plotting time series as part of time series and predictive modeling. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to plotting time series. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

03. Moving averages

Learning focus. This section introduces moving averages as part of time series and predictive modeling. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to moving averages. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

04. Missing values

Learning focus. This section introduces missing values as part of time series and predictive modeling. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to missing values. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

05. Serial correlation

Learning focus. This section introduces serial correlation as part of time series and predictive modeling. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to serial correlation. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

06. Autocorrelation

Learning focus. This section introduces autocorrelation as part of time series and predictive modeling. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to autocorrelation. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

07. Predictive modeling overview

Learning focus. This section introduces predictive modeling overview as part of time series and predictive modeling. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to predictive modeling overview. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

08. Train and test data

Learning focus. This section introduces train and test data as part of time series and predictive modeling. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to train and test data. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

09. Evaluating predictive models

Learning focus. This section introduces evaluating predictive models as part of time series and predictive modeling. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to evaluating predictive models. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

10. Regression metrics

Learning focus. This section introduces regression metrics as part of time series and predictive modeling. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to regression metrics. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

11. Classification metrics

Learning focus. This section introduces classification metrics as part of time series and predictive modeling. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to classification metrics. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

12. Building predictive solutions

Learning focus. This section introduces building predictive solutions as part of time series and predictive modeling. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to building predictive solutions. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

13. Sentiment analysis

Learning focus. This section introduces sentiment analysis as part of time series and predictive modeling. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to sentiment analysis. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

14. Module review

Learning focus. This section introduces module review as part of time series and predictive modeling. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to module review. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?